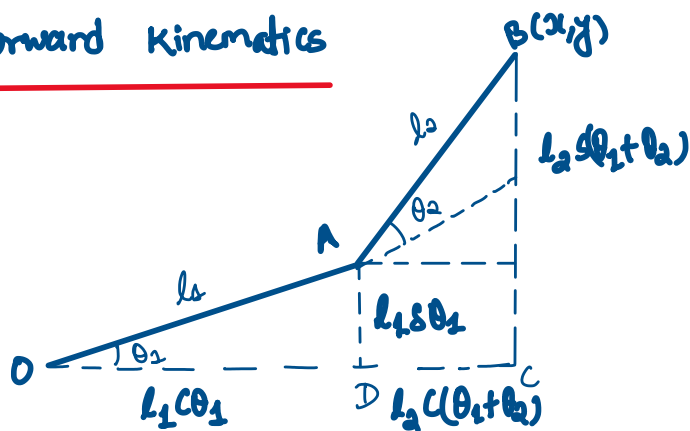


FK and IK for SCARA 2D

Wednesday, August 26, 2026

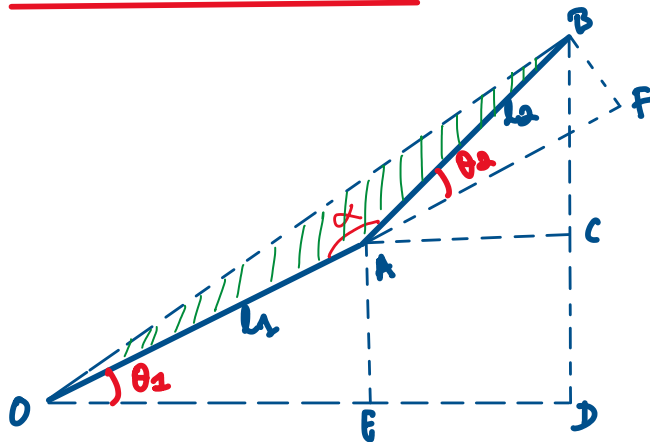
10:16 PM

① Forward Kinematics



$$\left. \begin{aligned} x &= l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2) \\ y &= l_1 \sin(\theta_1) + l_2 \sin(\theta_1 + \theta_2) \end{aligned} \right\}$$

② Inverse Kinematics:



Δ OAB :

$$r^2 = x^2 + y^2 = l_1^2 + l_2^2 - 2 \cdot l_1 \cdot l_2 \cdot \cos \alpha$$

$$\cos \alpha = \frac{l_1^2 + l_2^2 - r^2}{2 \cdot l_1 \cdot l_2}$$

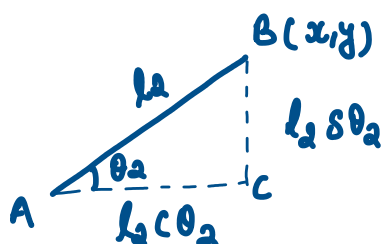
Now, $\theta = \pi - \alpha$

$$\cos \theta = \cos(\pi - \alpha) = -\cos \alpha$$

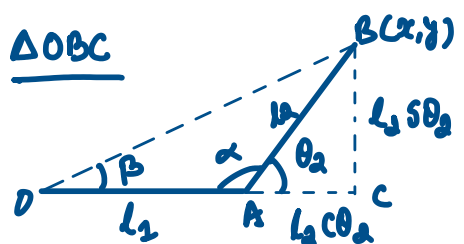
$$\cos \theta = \frac{r^2 - l_1^2 - l_2^2}{2 \cdot l_1 \cdot l_2} = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2 \cdot l_1 \cdot l_2}$$

$$\theta_2 = \cos^{-1} \left(\frac{x^2 + y^2 - l_1^2 - l_2^2}{2 \cdot l_1 \cdot l_2} \right)$$

Δ ABC

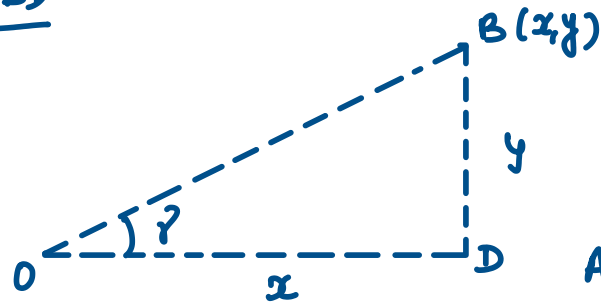


Δ OBC



$$\tan \beta = \frac{l_2 \sin \theta_2}{l_1 + l_2 \cos \theta_2} \Rightarrow \beta = \tan^{-1} \left(\frac{l_2 \sin \theta_2}{l_1 + l_2 \cos \theta_2} \right)$$

ΔOBD



$$\tan \gamma = \frac{y}{x}$$

$$\gamma = \tan^{-1} \frac{y}{x}$$

$$\text{Also, } \gamma = \theta_1 + \beta$$

$$\theta_1 = \gamma - \beta$$

$$\theta_1 = \tan^{-1} \frac{y}{x} - \tan^{-1} \left(\frac{l_2 \sin \theta_2}{l_1 + l_2 \cos \theta_2} \right)$$